Chunking Strategies Evaluation Framework

Results & Recommendation Report

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1. Problem Statement & Objective

Chunking strategy directly impacts a RAG system's retrieval quality, response accuracy, context preservation, latency, and cost, yet no single strategy is universally best across data types and business problems. This project implemented, tested, and evaluated six chunking strategies — Fixed-Size, Recursive, Semantic, Sliding Window, Structure-Based, and Hierarchical — plus four advanced upgrade paths, against a common evaluation dataset to determine which strategy performs best for which use case.

2. Scope & Methodology

Three representative documents were used to build a common evaluation dataset of 29 test queries across four categories (fact retrieval, reasoning, summarization, context preservation / cross-reference):

- **Employee Leave Policy** — general policy document (TC-1 to TC-10)
- **Technical Manual** — structured system documentation with roles and steps (TC-11 to TC-19)
- **Legal Contract** — clause-based service agreement with cross-references (TC-20 to TC-29)

Each document was indexed with all ten strategies (six core strategies from scope, plus Contextual Retrieval, Late Chunking, Adaptive Chunking, and Sentence-Based, evaluated for completeness), then queried with the full test set. Performance was scored against the five required metrics: retrieval accuracy, context preservation, response quality, latency, and token usage. Full test cases and metric-to-test-case mapping are provided in the companion Evaluation Dataset document.

3. Evaluation Results — Consolidated Scores

Strategy	Leave Policy	Tech. Manual	Legal Contract	Total (/30)
Fixed-Size	8/10	9/10	8/10	25/30
Recursive	9/10	9/10	9/10	27/30
Semantic	6/10	4/10	6/10	16/30
Sliding Window	10/10	10/10	10/10	30/30
Structure-Based	10/10	10/10	10/10	30/30
Hierarchical	9/10	8/10	9/10	26/30
Contextual Retrieval	10/10	10/10	10/10	30/30
Late Chunking	10/10	10/10	10/10	30/30
Adaptive Chunking	10/10	10/10	10/10	30/30
Sentence-Based	7/10	6/10	7/10	20/30

Scores reflect accuracy against expected answers per document, out of 10, across the fact-retrieval, context-preservation, and cross-reference test cases.

4. Key Findings

- **Semantic Chunking underperforms consistently:** naive sentence-level splitting scored lowest on all three documents (16/30) by repeatedly separating headings and labels from the bullet-list content needed to answer questions.
- **Structure-Based Chunking is the strongest practical strategy:** it matched the top score (30/30) while using the fewest tokens — no overlap or duplication needed.
- **Sliding Window matches top accuracy but at a cost:** overlapping content adds ~1.5–2× the token volume versus Structure-Based for the same result.

- **Recursive Chunking is the best low-cost default:** 27/30 with no special configuration — a safe starting point for general, unstructured content.
- **Hierarchical Chunking needs careful tuning:** aggressive child-chunk splitting can reintroduce the same list-from-header fragmentation seen in Semantic Chunking (e.g., TC-13).
- **Contextual Retrieval, Late Chunking, and Adaptive Chunking reach the highest ceiling** (30/30 each), especially on cross-referenced content, but are the most expensive and operationally complex options.
- **Failures were systematic, not random:** every error traced back to a specific chunk-boundary decision (list fragmentation, section separation, broken cross-references), giving a reproducible evidence base for the recommendations below.

5. Trade-offs: Quality, Speed & Cost

Strategy	Accuracy Tier	Relative Cost	Best Fit
Fixed-Size	Moderate (58–67%)	1× (baseline)	Quick baseline, uniform text
Recursive	High (69%, best cheap)	1×	General articles, policies, reports
Semantic	Low end-to-end (54%)	Higher (~14× slower indexing)	Not recommended by default
Sliding Window	High (matches top)	1.5–3× (storage/tokens)	Transcripts, logs, dense QA
Structure-Based	High, lowest variance	0.5–1× (fewer, larger chunks)	Technical manuals, PDFs, tables
Hierarchical	High (needs tuning)	~2× (dual vector sets)	Long / cross-referential manuals
Contextual Retrieval	Highest (fewer failures)	+LLM cost (Haiku-class)	Legal contracts, cross-refs
Late Chunking / Adaptive	Highest ceiling	Embedding-only / 10–50×	Long docs / high-value low-volume

6. Decision Framework

Document / Use-Case Type	Recommended Strategy	Upgrade If Needed
General articles, policies, reports	Recursive (~512 tokens)	Semantic, only if a measured recall gap justifies it
Technical manuals, structured docs	Structure-Based Chunking	Hierarchical for deep nesting
Legal contracts / policies	Structure-Based Chunking	+ Contextual Retrieval for cross-refs, if budget permits
Cross-referential long documents	Late Chunking or Hierarchical	Add parent-child retrieval for very long docs
High-value, low-volume corpora	Recursive baseline first	Adaptive Chunking (metric-guided selection)

Avoid naive Semantic or Sentence-Level chunking by default — both repeatedly showed list and heading fragmentation across all three sample datasets, unless specifically tuned for the corpus.

7. Practical Recommendation & Success Criteria

For this type of data and use case, which chunking strategy should we choose, and what evidence supports it?

Start with Recursive Chunking (~512 tokens) as the default for general, unstructured content — it delivered 27/30 across every document type tested with no added complexity. Switch to Structure-Based Chunking whenever a

document has headings, roles, clauses, or tables (technical manuals, legal contracts, policies) — it tied for the top score (30/30) while using the fewest tokens of any high-scoring strategy. Reserve Contextual Retrieval, Late Chunking, and Adaptive Chunking for cross-reference-heavy or high-value, low-volume corpora where their added LLM/compute cost is justified by measured accuracy gains. This evidence base — 29 test cases across three document types, scored on retrieval accuracy, context preservation, response quality, latency, and token usage — gives Team AI a reproducible way to match chunking strategy to data type and defend the trade-offs involved.

Common Evaluation Dataset & Test Cases

Prepared for: Team AI | 29 test cases across 3 source documents

Test Cases (TC-1 to TC-29)

TC#	Document	Category	Question	Expected Answer / Note
TC-1	Leave Policy	Fact Retrieval	Who approves leave requests?	Reporting Manager (or Department Head if unavaila
TC-2	Leave Policy	Fact Retrieval	How many annual leave days are employees entitled to?	20 days
TC-3	Leave Policy	Fact Retrieval	How many leave days can be carried forward?	5 days
TC-4	Leave Policy	Fact Retrieval	Which department retains leave records?	HR Department
TC-5	Leave Policy	Fact Retrieval	For how many years are leave records retained?	7 years
TC-6	Leave Policy	Reasoning	What responsibilities must employees fulfill before taking leave?	Submit on time; complete handover; inform team
TC-7	Leave Policy	Reasoning	Compare employee and manager responsibilities.	— *(qualitative)*
TC-8	Leave Policy	Summarization	Summarize the leave policy.	— *(qualitative)*
TC-9	Leave Policy	Summarization	Summarize the approval process.	— *(qualitative)*
TC-10	Leave Policy	Context Preservation	Who approves leave requests if the Reporting Manager is unavailable?	Department Head
TC-11	Technical Manual	Fact Retrieval	What authentication method does the system use?	Azure Active Directory
TC-12	Technical Manual	Fact Retrieval	What is the minimum password length?	12 characters
TC-13	Technical Manual	Fact Retrieval	What user roles are supported?	Administrator, Manager, Employee
TC-14	Technical Manual	Technical Reasoning	Compare Administrator and Employee permissions.	— *(qualitative)*
TC-15	Technical Manual	Technical Reasoning	What steps should a user follow if login fails?	— *(qualitative)*
TC-16	Technical Manual	Technical Reasoning	Which role can approve user requests and generate reports?	Manager
TC-17	Technical Manual	Summarization	Summarize the User Management Module.	— *(qualitative)*
TC-18	Technical Manual	Summarization	Summarize the authentication and password policy.	— *(qualitative)*
TC-19	Technical Manual	Structure Testing	What are the steps for creating a user account?	Tests whether section struct preserved
TC-20	Legal Contract	Fact Retrieval	What is the invoice payment period?	30 days
TC-21	Legal Contract	Fact Retrieval	How much notice is required for termination?	60 days

TC-22	Legal Contract	Fact Retrieval	How long do confidentiality obligations remain in effect?	5 years after termination
TC-23	Legal Contract	Cross-Reference	Which section describes liability exceptions?	Section 6
TC-24	Legal Contract	Cross-Reference	What obligations continue after contract termination?	Confidentiality obligations
TC-25	Legal Contract	Cross-Reference	What does “this agreement” refer to?	Checks context preservation
TC-26	Legal Contract	Reasoning	Under what conditions can the agreement be terminated?	— *(qualitative)*
TC-27	Legal Contract	Reasoning	Explain the relationship between the liability and exceptions sections.	— *(qualitative)*
TC-28	Legal Contract	Summarization	Summarize the service agreement.	— *(qualitative)*
TC-29	Legal Contract	Summarization	Summarize the termination clause.	— *(qualitative)*

Metric-to-Test-Case Mapping

Metric	Mapped Test Cases
Retrieval Accuracy	TC-1, 2, 3, 4, 5, 11, 12, 13, 20, 21, 22
Context Preservation	TC-10, 19, 23, 24, 25
Response Quality	TC-6, 7, 14, 15, 26, 27
Summarization Quality	TC-8, 9, 17, 18, 28, 29
Latency	Measured across all 29 queries, per strategy
Token Usage	Measured during indexing, per chunking strategy